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Original Research Article

Xiaoyao San, a Chinese herbal formula, ameliorates depression-like behavior in mice through the AdipoR1/AMPK/ACC pathway in hypothalamus

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ABSTRACT

Objective: Depression and metabolic disorders have overlapping psychosocial and pathophysiological causes. Current research is focused on the possible role of adiponectin in regulating common biological mechanisms. Xiaoyao San (XYS), a classic Chinese medicine compound, has been widely used in the treatment of depression and can alleviate metabolic disorders such as lipid or glucose metabolism disorders. However, the ability of XYS to ameliorate depression-like behavior as well as metabolic dysfunction in mice and the underlying mechanisms are unclear.
Methods: An *in vivo* animal model of depression was established by chronic social defeat stress (CSDS). XYS and fluoxetine were administered by gavage to the drug intervention group. Depression-like behaviors were analyzed by the social interaction test, open field test, forced swim test, and elevated plus maze test. Glucose levels were measured using the oral glucose tolerance test. The involvement of certain molecules was validated by immunofluorescence, histopathology, and Western blotting. *In vitro*, hypothalamic primary neurons were exposed to high glucose to induce neuronal damage, and the neuroprotective effect of XYS was evaluated by cell counting kit-8 assay. Immunofluorescence and Western blotting were used to evaluate the influences of XYS on adiponectin receptor 1 (AdipoR1), adenosine 5′-monophosphate-activated protein kinase (AMPK), acetyl-coenzyme A carboxylase (ACC) and other related proteins.
Results: XYS ameliorated CSDS-induced depression-like behaviors and glucose tolerance impairment in mice and increased the level of serum adiponectin. XYS also restored Nissl bodies in hypothalamic neurons in mice that exhibited depression-like behaviors and decreased the degree of neuronal morphological damage. *In vivo* and *in vitro* studies indicated that XYS increased the expression of AdipoR1 in hypothalamic neurons.
Conclusion: Adiponectin may be a key regulator linking depression and metabolic disorders; regulation of the hypothalamic AdipoR1/AMPK/ACC pathway plays an important role in treatment of depression by XYS.

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1. Introduction

Depression is a common mental disorder mainly characterized by significant and persistent depressive symptoms, which can seriously affect patients' thinking, cognition, and physical function [1]. In the past two years, the global coronavirus disease-2019 pan-

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demic has posed a significant threat to public mental health worldwide, and the prevalence of depression has increased year by year, with an average prevalence of 34.31% [2,3]. Current antidepressant drugs, specifically selective serotonin reuptake inhibitors, mainly act by regulating the monoaminergic system [4]. However, approximately 30% of depression patients are partially or completely resistant to treatment [5]. Thus, new, safe and effective drugs for complex depressive symptoms are urgently needed.

Convincing evidence indicates that depression is strongly associated with metabolic disorders, including dyslipidemia, obesity, type 2 diabetes, and insulin resistance [6]. In the clinic, depression and diabetes occur together approximately twice as frequently as would be predicted by chance alone [7,8], which means that the risk of metabolic complications is significantly increased in individuals with depression. Adipose tissue, an active metabolic and endocrine organ, responds to environmental and psychological challenges involving complex interactions between the nervous system and the endocrine system [9]. Adipocytes can manufacture and secrete over 600 bioactive factors, called adipokines [10]. Some adipokines can act on the nervous system and mediate crosstalk between adipose tissue and the brain by regulating neuronal excitability and synaptic plasticity and promoting neural circuit remodeling.

Adiponectin is the most abundant adipokine in the human blood, with a physiological concentration of 5–30 μg/mL [11], and it has insulin-sensitizing, anti-inflammatory and antioxidant effects [12]. The function of adiponectin in lipid regulation and energy metabolism mainly depends on the receptor-ligand type interaction, in which adiponectin binds to its cognate receptor and initiates multiple intracellular signaling mechanisms through the adenosine 5′-monophosphate-activated protein kinase (AMPK), mammalian target of rapamycin, and c-Jun *N*-terminal kinase pathways [13]. Several studies have shown that adiponectin has a wider range of effects and functions in the brain than previously thought and that lower levels of adiponectin are associated with major depressive disorder [14,15].

Here, we show that Xiaoyao San (XYS), a famous traditional Chinese medicine compound with a long history of clinical use, effectively alleviates depression-like behaviors in a mouse model of depression induced by chronic social defeat stress (CSDS) [16,17]. We speculate that adiponectin may be a key regulator of depression and glucose metabolism disorders, and XYS may improve glucose tolerance and depression-like behaviors in mice by regulating the adiponectin receptor 1 (AdipoR1)/AMPK/acetyl-coenzyme A carboxylase (ACC) pathway in the hypothalamus

2. Materials and methods

2.1. Animal grouping

Adult male specific pathogen-free C57BL/6J mice (8 weeks old) were purchased from SiPeiFu Biotechnology Co., ltd. (Beijing). The study was carried out in the Laboratory of the Formula-Pattern of Jinan University (Guangzhou, China) and strictly followed the relevant experimental procedures. The animals were maintained at a controlled temperature ([20 ± 2] °C) and humidity (60% ± 10%) on a 12-hour-light/12-hour-dark cycle. After 1 week of adaptive feeding, the mice were randomly divided into the following four groups according to weight (mean weight = 22.6 g): Control group, CSDS group, CSDS + XYS group, and CSDS + fluoxetine (FLX) group (n = 8 per group). After the start of the experiment, changes in blood glucose levels in each group of mice were measured on days 3, 6 and 9. This study was approved by the Animal Experiment Ethics Committee of Jinan University (ethics number: 20211216–06), and the animal protocol was performed in accordance with the appropriate guidelines.

2.2. Drug preparation and administration

XYS is composed of 8 Chinese herbal medicines, including Chaihu (*Radix Bupleuri*), Danggui (*Radix Angelicae Sinensis*), Baishao (*Radix Paeoniae Alba*), Cangshu (*Rhizoma Atractylodis Macrocephalae*), Fuling (*Poria*), Gancao (*Radix Glycyrrhizae*), Bohe (*Herba Menthae*) and Shengjiang (*Rhizoma Zingiberis Recens*), in a 6:6:6:6:6:3:2:2 ratio. All herbal medicines were purchased from Jiuzhitang Co., Ltd. (Changsha, China). In a previous study [18], we quantified the content of the active ingredients in XYS using ultra-high-performance liquid chromatography-quadrupole time-of-flight mass spectrometry technique. For *in vivo* experiments, XYS was dissolved in deionized water, and mice were administered XYS intragastrically at a dosage of 0.3 g/kg per day [19,20].

To prepare freeze-dried XYS powder, pure water was added to 200 g of the XYS powder mentioned above to a volume of 100 mL. The concentrated decoction was mixed in a sterile glass bottle so that the quality of sample in each tube was uniform. The mixtures were frozen according to the following gradient cooling method: frozen in a −20 °C freezer for 4 h and then frozen in a −80 °C freezer for 24 h. Finally, using vacuum freeze-drying technology, the freeze-dried powder was collected and stored in a −20 °C freezer. The freeze-dried powder was identified by experts at the School of Pharmacy, Jinan University. For *in vitro* experiments, freeze-dried XYS powder was diluted to various concentrations using Dulbecco's modified Eagle medium (DMEM, Gibco, USA).

FLX was purchased from Macklin Inc. (China) and was dissolved in deionized water. FLX was administered to the appropriate group of mice at a dosage of 3.0 mg/kg per day [21]. Mice in the control group and CSDS group were gavaged with an equal volume of deionized water.

Recombinant mouse globular adiponectin (ProSpec-TechnoGene ltd., Israel) was diluted to a concentration of 1.0 μg/mL in DMEM.

2.3. CSDS

The day before modeling, CD1 mice were placed on one side of the cage for acclimation. During the 10-day modeling period, C57BL/6 mice were exposed to CD1 mice for 10 min every day. During the last 24 h, CD1 mice and C57BL/6 mice were placed on either sides of the same cage and separated by perforated transparent partitions, which allowed visual, olfactory and auditory contact within 24 h of social failure. Behavioral tests were performed 10 days after modeling [22].

2.4. Behavioral tests

2.4.1. Social interaction test

Social interaction test was conducted in a social interaction box (42 cm × 42 cm × 50 cm, Noldus Information Technology, Netherlands) 24 h after the last social failure. The box comprised an interaction area with a mesh target box (10 cm × 4.5 cm × 10 cm) and two diagonal areas. The test was divided into two phases: the no-social target and social target phases. In the no-social target phase, CD1 mice were not placed in the mesh target box, and experimental mice were placed in the box and allowed to move freely for 2.5 min. After this phase, CD1 mice were placed in the mesh target box, and the experimental mice were again placed in the box and allowed to move freely for 2.5 min. EthoVision 14.0 software (Noldus Information Technology, Netherlands) was used to record and measure the amount of time spent in the interaction area and the ratio of time spent with social targets to the time spent without social targets.

2.4.2. Open field test

Open field test was used to analyze general locomotion and exploratory behavior. All animals were individually placed in an open field box (50 cm × 50 cm × 40 cm, white bottom) and allowed to explore freely for 5 min for adaptation. The activity of all mice was recorded with a camera mounted above the box. EthoVision 14.0 software was used to record and analyze the total distance traveled by the mice in the open field box and the time spent in the central area.

2.4.3. Forced swim test

In the forced swim test, the mice were individually placed in a transparent plastic cylinder (50 cm high, 20 cm diameter) filled with water (30 cm deep, 20–25 °C), and allowed to swim for 5 min. EthoVision 14.0 software was used to record and analyze the time spent floating in the water without struggling or exercising.

2.4.4. Elevated plus maze test

In the elevated plus maze test, the experimental animals were placed in the central area of the maze facing the arms. Each experimental animal was placed in the same position and allowed to explore freely for 5 min. EthoVision 14.0 software was used to record the cumulative time of the mouse entering the open arms or the closed arms, and to calculate the percentage of time into the open arms.

2.5. Sample collection

Serum and tissue samples were collected at 5 p.m. after the behavioral tests. The mice were placed in the anesthesia induction box of a small animal anesthesia machine (RWD Life Science Co., Ltd., Shenzhen, China). The concentration of isoflurane used for anesthesia induction was 4%. After the mice were deeply anesthetized, the eyeballs of the mice were quickly removed with surgical curved forceps, blood was collected in a sterile centrifuge tube, and the mice were killed by cervical dislocation immediately after blood collection. The blood samples were stored at 4 °C overnight, centrifuged at 3000 r/min for 10 min at 4 °C and stored in a −80 °C freezer. Ophthalmic forceps and ophthalmic scissors were used to quickly isolate liver tissues, perirenal adipose tissues and hypothalamic tissues, which were placed in cryopreservation tubes, and stored in a −80 °C freezer for analysis.

2.6. Isolation and culture of primary neurons

Primary hypothalamic neurons were obtained from neonatal rats within 24 h of birth. In short, neonatal rats were disinfected and placed in a petri dish containing alcohol on an ice box for ice anesthesia until the neonates showed no obvious movement. They were quickly decapitated, the brain was removed, and the hypothalamus was isolated and cut into small pieces. The tissue was digested, filtered, and centrifuged in trypsin (Gibco, USA). The supernatant was discarded, and the cells were resuspended in neurobasal medium (Gibco, USA) supplemented with 2% B27 (Thermo Fisher Scientific, USA), penicillin (20 U/mL) and streptomycin (20 mg/mL) (Gibco, USA) [23]. The cells were counted and cultured in a 5% CO_2 incubator at 37 °C. Then, the medium was replaced at appropriate times according to the cell conditions.

2.7. Analysis of cell viability and apoptosis

2.7.1. Cell viability detection

Cells in the control group were grown in DMEM containing 5.56 mmol/L glucose, and cells in the high-glucose treatment group were grown in DMEM containing 100 mmol/L glucose [24]. Afterward, cells in the high-glucose treatment group were incubated with different concentrations of XYS for 24 h. Then, cell viability was measured using the cell counting kit-8 (CCK-8, Bestbio, China) assay. Briefly, 5000 cells were added to each well of a 96-well plate for culture and drug treatment. Then, 10 μL of CCK-8 solution was added to each well, and the cells were incubated for 1 h. The absorbance was measured at 450 nm using the microplate reader (Epoch2, Bio Tek Instruments, USA).

2.7.2. Cell apoptosis detection

Cells on slides were fixed in 4% paraformaldehyde for 30 min. Next, a solution containing terminal deoxynucleotidyl transferase and fluorescein-labeled 2′-deoxyuridine 5′-triphosphate (dUTP) was dropped onto the sections, and the sections were incubated in the dark. A total of 50 μL terminal deoxynucleotidyl transferase dUTP nick end labeling (TUNEL) reagent (Beyotime Biotechnology Co., China) was added to each slice, and the cells were incubated at 37 °C for 60 min. Nuclear staining reagents and sealants were used according to instructions from the TUNEL kit manufacturers. After staining, cells in five randomly selected fields of view were observed under a fluorescence microscope (Olympus Corporation, Japan). The numbers of 4′,6-diamidino-2-phenylindole (DAPI)-positive and TUNEL-positive cells in each field were determined using ImageJ software (Version 1.53c, National Institutes of Health, USA). The percentage of TUNEL-positive cells was calculated using the following formula: TUNEL-positive cells (%) = TUNEL-positive cells/DAPI-positive cells × 100% [25].

2.8. Oral glucose tolerance test

Blood was obtained from the tails of mice. To prevent stress-induced hyperglycemia, the mice were acclimated to tail pinches for 1 week before the formal oral glucose tolerance test (OGTT). On the day before the experiment, the mice were placed in a cage with clean litter and fasted for 14 h. The mice in each group were given 1 g/kg glucose (20% aqueous solution) by oral gavage, and the blood glucose levels were measured at 0, 15, 30, 60 and 120 min. The tail tip (1–1.5 mm) of each mouse was cut off, the first drop of blood was discarded, and a blood glucose meter (Johnson & Johnson Services, Inc., USA) was used to measure the blood glucose content in the second drop of blood.

2.9. Measurement of corticosterone, leptin and adiponectin levels

The levels of corticosterone (MU30431), leptin (MU30381) and adiponectin (MU30297) were measured by enzyme-linked immunosorbent assay (all kits were purchased from Bioswamp Life Science Lab, China). The experiments were performed according to the instructions of the commercial kits. A sample or standard (50 μL) was directly added to each well, and then 50 μL of biotinylated antigen was added to all the wells except the blank. The plate was incubated at 37 °C for 60 min. Then, the plate was washed, 50 μL of enzyme-labeled avidin was added to each well except the blank control well, and the plate was incubated at 37 °C for 30 min. Then, 50 μL of termination liquid was added to each well. A microplate reader (Epoch2, Bio Tek Instruments, USA) was used to measure the absorbance at a wavelength of 450 nm.

2.10. Histopathological examination

Liver tissues, perirenal adipose tissues and hypothalamic tissues were fixed with paraformaldehyde, dehydrated in graded ethanol solutions, cleared with xylene, and finally embedded in paraffin. The tissues were sliced using a tissue microtome (Leica Biosystems, Germany), and the tissues were subjected to hematoxylin-eosin (HE) staining and Nissl staining with an HE staining kit (Beyotime Biotechnology Co., China) and a Nissl staining kit (Beyotime Biotechnology Co., China), respectively. Images were obtained using a fluorescence microscope (Olympus Corporation, Japan).

2.11. Immunofluorescence

Brain slices or cells were fixed, treated with blocking buffer, and then incubated overnight with antibodies against adiponectin receptor 1 (AdipoR1) (1:200, Abcam, UK) or microtubule-associated protein 2 (Map-2, 1:200, Cell Signaling Technology, USA) at 4 °C. After washing with phosphate buffer saline (PBS) three times, the slices or cells were incubated with the corresponding fluorescent-conjugated secondary antibody for 1 hour at room temperature. Then, the samples were washed three times with PBS and incubated with DAPI (Thermo Fisher Scientific, USA) for 5 min at room temperature. Images were obtained using confocal microscopy (Leica Biosystems, Germany).

2.12. Western blotting analysis

Total protein was extracted from hypothalamic tissue or primary neurons with radio-immunoprecipitation assay lysis buffer (Beyotime Biotechnology Co., China) containing 1% protease inhibitor cocktail (Beyotime Biotechnology Co., China) and 1% phosphatase inhibitor (Merck Millipore, Germany). The protein concentration in the lysate was determined using the bicinchoninic acid protein assay kit (Thermo Fisher Scientific, USA). A total of 30 μg of protein was loaded on a 10% sodium dodecyl sulfate polyacrylamide gel, and then the resolved proteins were transferred to a polyvinylidene fluoride membrane using a standard wet transfer system (Bio-Rad Laboratories, Inc., USA). The membrane was blocked with 5% skim milk at room temperature for 1 h and then incubated with the following primary antibodies overnight at 4 °C: AdipoR1 (1:1000, Abcam, ab126611, UK), phospho-adenosine 5′-monophosphate-activated protein kinase (p-AMPK, 1:1000, Cell Signaling Technology, #8208, USA), AMPK (1:1000, Cell Signaling Technology, #2532, USA), phospho-acetyl-coenzyme A carboxylase (p-ACC, 1:1000, Cell Signaling Technology, #11818, USA), and ACC (1:1000, Cell Signaling Technology, #3676, USA). Subsequently, the membrane was washed with Tris-buffered saline with Tween 20 (TBST) three times for 10 min each. The membranes were incubated with the appropriate horseradish peroxidase-labeled secondary antibody at room temperature for 1 h and then rinsed again with TBST three times for 10 min each. The protein bands were visualized with the ChemiDoc™ Touch Imaging System (Bio-Rad Laboratories, Inc., USA).

2.13. Statistical analysis

All data were analyzed with SPSS 20.0 software (IBM SPSS Statistics, USA) and expressed as the mean ± standard error of the mean. OGTT data were analyzed using repeated-measures analysis of variance (ANOVA) to determine significant differences. Other data were analyzed using one-way ANOVA followed by Dunnett's test. A P value < 0.05 was considered a statistically significant difference. All figures were plotted in GraphPad Prism 7.0 Software (GraphPad Software, USA).

3. Results

3.1. XYS alleviated CSDS-induced depression-like behavior in mice

To evaluate the antidepressant effect and mechanism of XYS, we exposed male C57BL6/J mice to CSDS (Fig. 1A), which induces a variety of depression-like symptoms, including a social avoidant phenotype in stress-susceptible mice. In the social interaction test, when CD1 mice were present in the mesh target box, mice in the CSDS group stayed in the social interaction area for a significantly shorter amount of time than mice in the control group ($P < 0.01$), indicating obvious social escape behavior (Fig. 1B). After intervention with XYS or positive control drugs, the model mice stayed in the interaction area for a significantly longer amount of time ($P < 0.05$). In the forced swim test, XYS reduced the immobility time of the model mice (Fig. 1C). The total distances traveled in the open field test and elevated plus maze test were not significantly different between the model mice and the control mice, indicating that the motor ability of the mice in the different groups was not abnormal. Time to stay in center area and open arms was lower in the CSDS group than in the control group, and mice in the CSDS group exhibited reduced active exploration behavior, indicating that CSDS caused anxiety-like behaviors in mice (Fig. 1D and E). After XYS intervention, the latency to enter the central area and open arms increased significantly. Behavioral testing showed that XYS and the positive control drugs alleviated CSDS-induced depression-like behavior in mice.

3.2. XYS ameliorated metabolic dysfunction in mice with depression-like behavior

Dysregulation of the hypothalamus-pituitary-adrenal (HPA) axis and other systemic metabolic disorders are closely related to depression. For this reason, we monitored the changes in blood glucose levels in each group of mice during the modeling period and the changes in glucose tolerance after modeling (Fig. 2A and B). The results showed that the blood glucose levels of the mice in CSDS group increased significantly during the modeling period compared with that in the control group ($P < 0.01$). According to the OGTT, mice in the CSDS group exhibited impaired glucose tolerance. After intervention with XYS, the glucose tolerance of the model mice was significantly improved ($P < 0.01$).

Glucocorticoids have a major impact on the metabolism of carbohydrates, lipids and proteins. The blood corticosterone level of the CSDS group was significantly higher than that of the control group, and XYS slightly reduced the glucocorticoid level. Unexpectedly, there was no significant difference in the level of either adiponectin or leptin in adipose tissue ($P > 0.05$). The level of adiponectin in the CSDS group was significantly lower than that in the control group ($P < 0.01$), which could be reversed by the administration of XYS or FLX ($P < 0.01$). This result suggests that serum circulating adiponectin may play an important role in the therapeutic effect of XYS.

3.3. Effects of XYS on the morphology of hypothalamic neurons in mice with depression-like behavior

The HPA axis is activated through the secretion of corticotropin-releasing hormone and vasopressin from the hypothalamus. In this study, we used HE staining and Nissl staining to observe the pathological changes in the hypothalamus in each group of mice.

HE staining revealed no obvious pathological changes (Fig. 3A). The distribution of hypothalamic neurons in each group of mice was relatively uniform, and the nuclei did not shrink or disappear.

Following Nissl staining, the number of Nissl bodies in the hypothalamus was significantly reduced in the CSDS group compared with the control group, and the overall blue staining degree

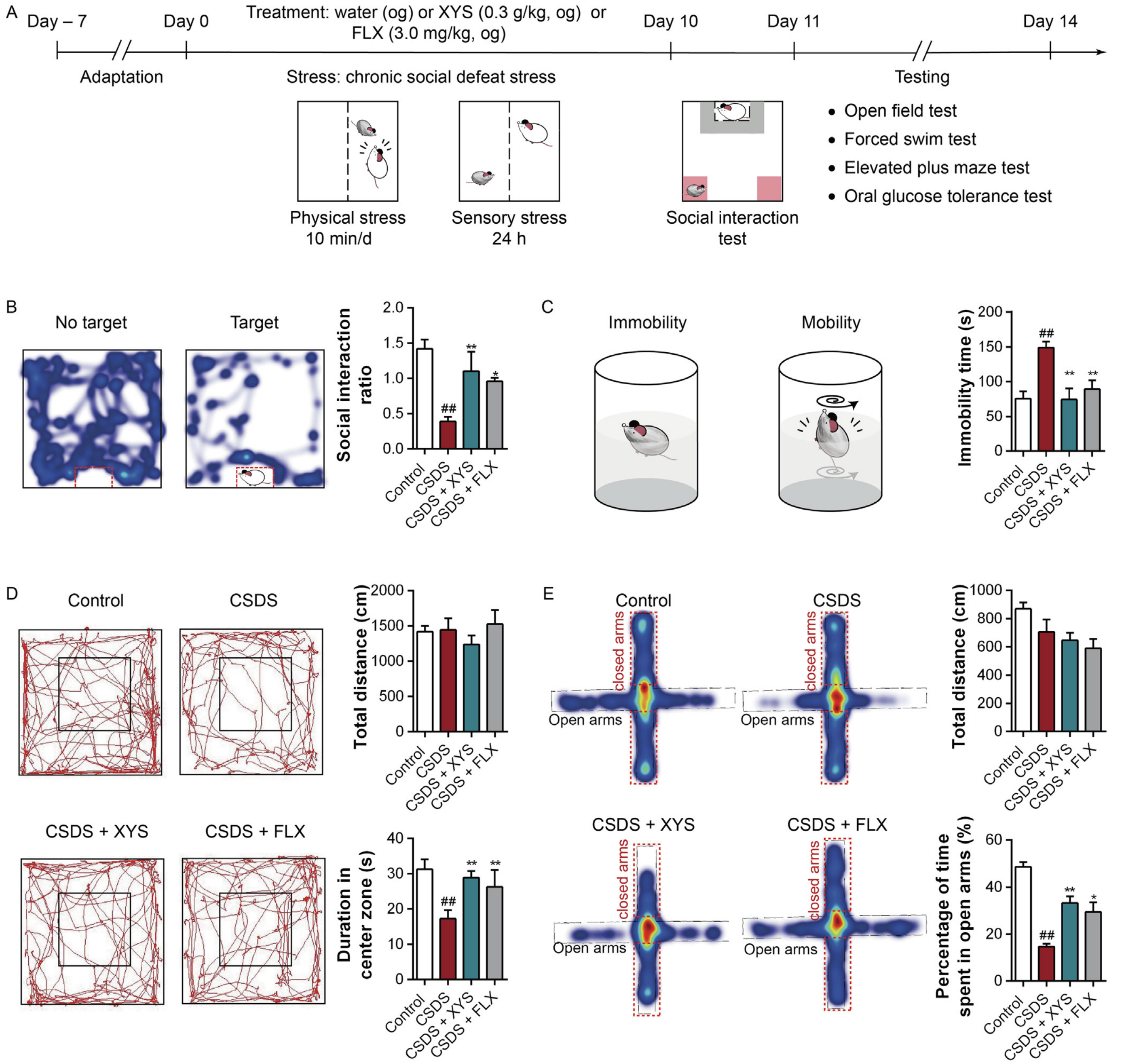

Fig. 1. Antidepressant effects of XYS. (A) Schematic diagram of the experimental process. (B) Social interaction test: heatmap of the trajectories of the mice in the interaction area in the presence and absence of CD1 mice (left), and the interaction ratio of mice in the interaction area when CD1 mice were present ($P = 0.0009$ within groups, $n = 8$) (right). (C) Forced swim test: schematic diagram of the forced swim test (left), and histogram of the immobility time of mice ($P = 0.0004$ within groups, $n = 8$) (right). (D) Open field test: representative images of mouse trajectory in the open field test (left), and the total movement distance of the mouse in the open field box ($P = 0.5647$ within groups, $n = 8$) and the cumulative time of the mouse in the central area ($P = 0.0253$ within groups, $n = 8$) (right). (E) Elevated plus maze: heatmap of mouse trajectory in the elevated plus maze (left), and the total distance traveled in the elevated plus maze ($P = 0.0288$ within groups, $n = 8$) and the percentage of time spent in open arms ($P = 0.0001$ within groups, $n = 8$) (right). $^{\#\#}P < 0.01$, *vs* control; $^{*}P < 0.05$, $^{**}P < 0.01$, *vs* CSDS. Data were analyzed by one-way analysis of variance followed by Dunnett's test. CSDS: chronic social defeat stress; FLX: fluoxetine; og: oral gavage; XYS: Xiaoyao San.

of the cells was low (Fig. 3B). Hypothalamic neural cells were significantly stimulated in the CSDS group, and XYS reduced the stimulation of hypothalamic neural cells ($P < 0.05$). Surprisingly, the number of Nissl bodies in hypothalamic neurons was not significantly increased in the FLX group compared with the CSDS group. Next, we simultaneously observed the pathological changes in adipose and liver tissues, two tissues closely related to metabolism. However, perhaps due to the short modeling time, HE staining revealed no obvious pathological changes in liver and adipose tissues (Fig. 3A). In all groups, hepatocytes were arranged regularly and showed no obvious swelling or necrosis; similarly, the adipocyte bodies showed large lipid droplets, and the cell boundaries were clear. In general, mice in the CSDS group exhibited depression-like behaviors and metabolic abnormalities, and the pathological changes in these mice mainly manifested as a decrease in the number of Nissl bodies in hypothalamic neural cells. However, XYS relieved these pathological changes.

3.4. Effects of XYS on the hypothalamic AdipoR1/AMPK/ACC pathway in mice with depression-like behavior

Next, we investigated the expression of AdipoR1 in the hypothalamic region. Through immunofluorescence staining, we found that the fluorescence intensity of AdipoR1 in the hypothala-

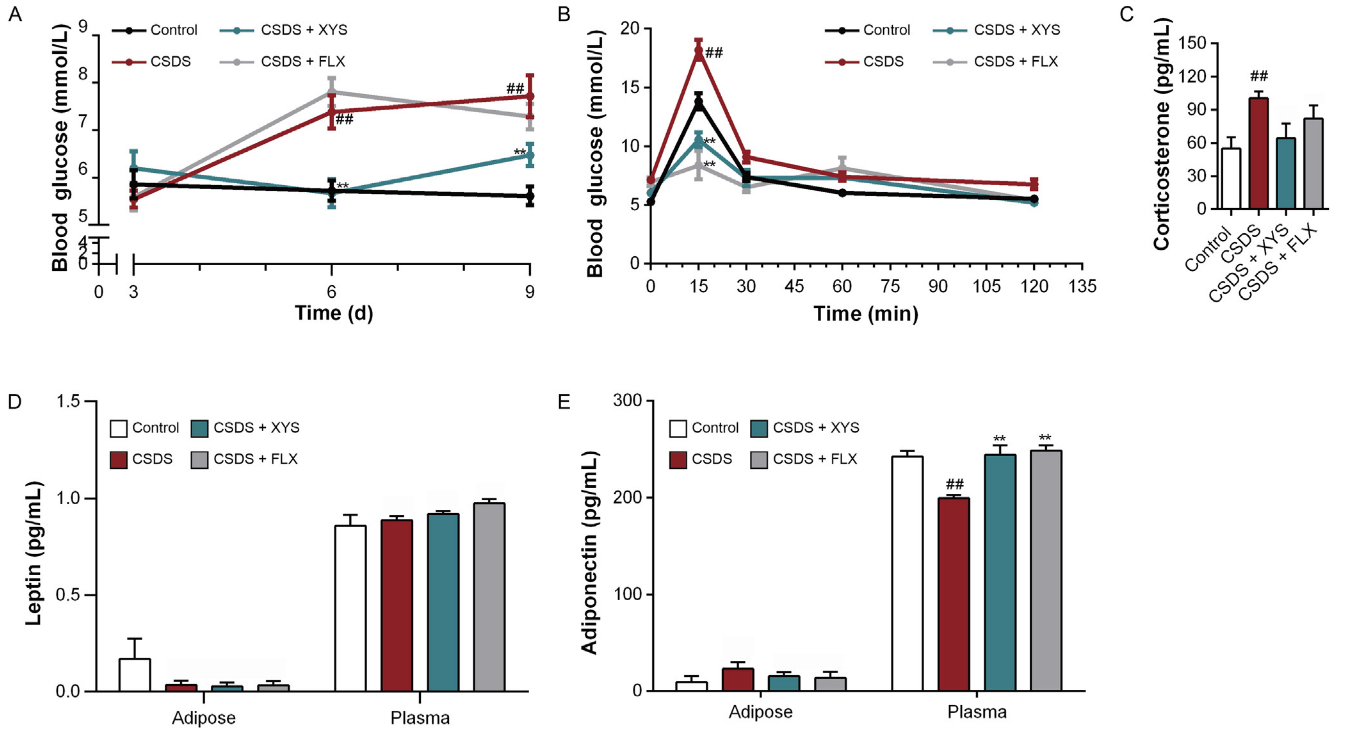

Fig. 2. Effects of XYS on mouse metabolism. (A) Blood glucose measured on the 3rd, 6th and 9th days during the induction of depression by CSDS in mice (n = 8). (B) Blood glucose measured at 0, 15, 30, 60 and 120 min (n = 8) for oral glucose tolerance test. (C) Serum corticosterone (P = 0.0502 within groups, n = 6). (D) Leptin levels in the adipose tissue (P = 0.1785 within groups, n = 6) and plasma (P = 0.1186 within groups, n = 6). (E) Adiponectin levels in the adipose tissue (P = 0.4420 within groups, n = 6) and plasma (P = 0.0004 within groups, n = 6). $^{\#\#}P$ < 0.01, *vs* control; $^{**}P$ < 0.01, *vs* CSDS. Data were analyzed by one-way analysis of variance followed by Dunnett's test. CSDS: chronic social defeat stress; FLX: fluoxetine; XYS: Xiaoyao San.

mus was significantly lower in the CSDS group than in the control group, and XYS increased the expression of AdipoR1 in the model mice. Hypothalamic neural cells in the CSDS group were significantly stimulated, and XYS reduced the stimulation (P < 0.05) (Fig. 4A and B). Then, Western blotting was performed for verification, and the results were consistent with the results of immunofluorescence staining (Fig. 4C). Studies have shown that adiponectin can regulate the oxidation and synthesis of fatty acids through the AdipoR1/AMPK/ACC signaling pathway to maintain cell energy balance. The immunoblotting results showed that the level of p-AMPK in the CSDS group was significantly reduced compared with that in the control group. AMPK was expressed at significantly higher level in the group treated with XYS than in the control group (P < 0.05). Moreover, the phosphorylation of ACC in the CSDS group was significantly lower than that in the control group, indicating that ACC was activated. After XYS intervention, the expression of p-ACC in the hypothalamus increased, leading to inhibition of lipid synthesis (Fig. 4D). These findings indicate that XYS can inhibit lipid synthesis in hypothalamic neural cells and reduce energy consumption, which is mainly inactivated via phosphorylation of ACC by AMPK.

3.5. Effects of XYS on primary neuron damage induced by high glucose

The above results showed that the CSDS-induced depression model mice had abnormally elevated blood glucose levels and impaired glucose tolerance during the modeling period. To clarify the mechanism underlying the regulatory effect of XYS on the AdipoR1/AMPK/ACC signaling pathway, we first employed *in vitro* experiments. We used high-glucose medium to induce mouse primary neuronal damage and treated the cells with freeze-dried XYS powder. The results showed that treatment with 0.1 mg/mL freeze-dried XYS powder improved the cell survival rate and had a protective effect against cell damage induced by high glucose (P < 0.05) (Fig. 5A). We found that in the high glucose-induced cell damage model, the apoptosis rate of cells was significantly higher than that in the control group. Adiponectin administration reduced the apoptosis rate of cells (P < 0.05) (Fig. 5B and C). Interestingly, the apoptosis rate of cells was only significantly decreased in the group treated simultaneously with freeze-dried XYS powder and adiponectin compared with the high glucose group, and the rate of decrease was higher than that in the adiponectin group. Freeze-dried XYS powder alone had no significant antiapoptotic effect (Fig. 5B and C). The results of the *in vitro* experiments suggest that freeze-dried XYS powder may not have an adiponectin-like effect but may increase the sensitivity of cells to adiponectin.

3.6. XYS regulated the AdipoR1/AMPK/ACC pathway in primary neurons following exposure to high glucose

Next, to explore whether XYS can increase sensitivity to adiponectin through the AdipoR1/AMPK/ACC signaling pathway *in vitro*, we performed immunofluorescence staining for a marker of primary neurons (Map-2) and AdipoR1. The fluorescence intensity of AdipoR1 in the high glucose group was significantly lower than that in the control group. The fluorescence intensity of AdipoR1 in neurons was significantly higher in the XYS treatment group than in the high glucose group (P < 0.05) (Fig. 6A and B). The Western blotting results were consistent with the immunofluorescence results (Fig. 6C). The expression level of p-AMPK, which is involved in the AMPK/ACC signaling pathway, in the neurons was significantly lower in the high glucose group than in the control group, indicating that neuronal AMPK was not activated in this group (Fig. 6D). It is worth noting that, the expression levels of p-AMPK and p-ACC in neurons were not increased in the XYS treatment group compared with the high glucose group (Fig. 6D). According to these results, XYS may increase sensitivity to adiponectin by

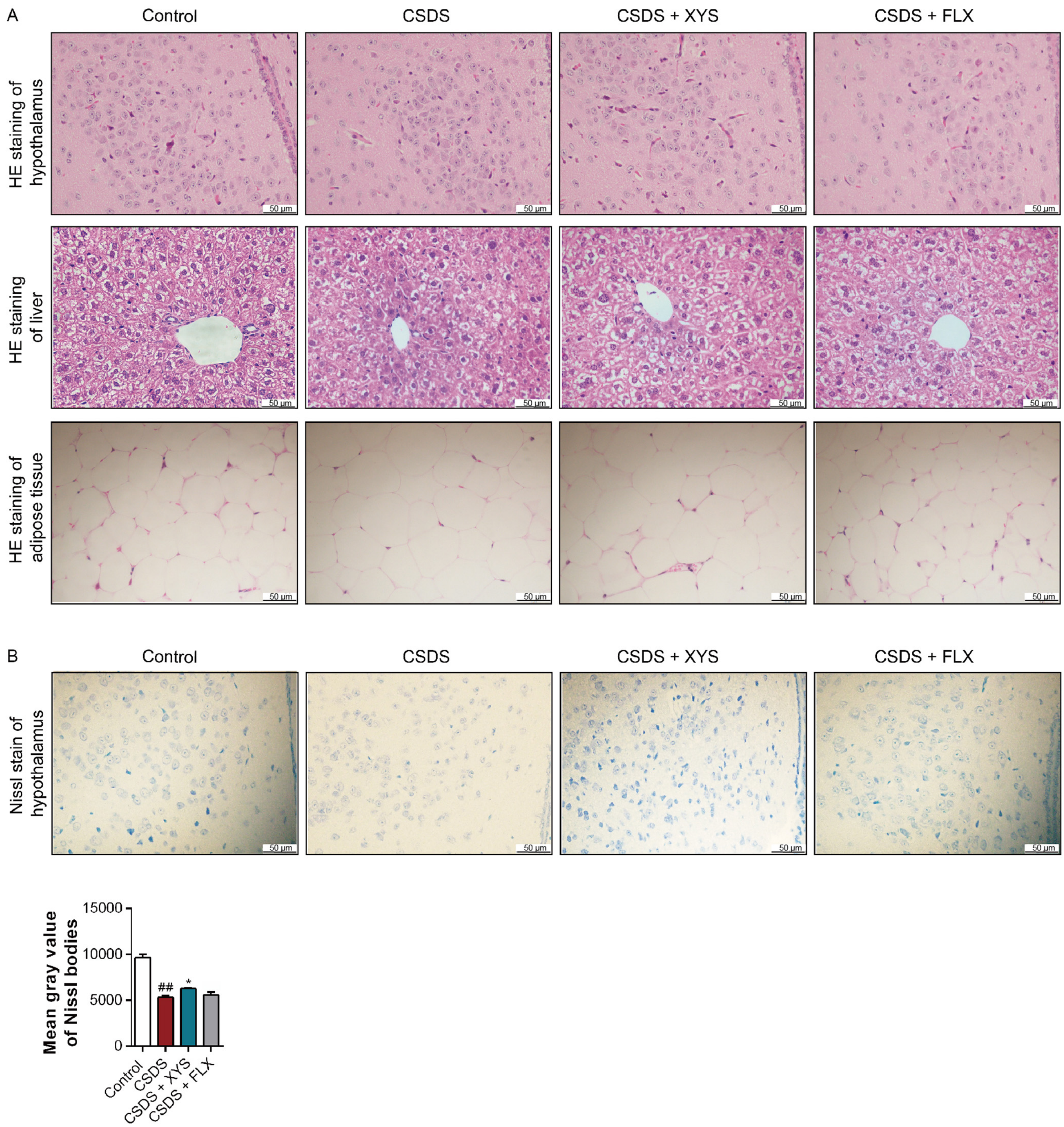

Fig. 3. Pathological changes in mouse tissues from each group. (A) HE staining of hypothalamic, liver and adipose tissues from mice in each group. (B) Nissl staining of the hypothalamus and grayscale statistics of Nissl bodies in each group of mice ($P = 0.0001$ within groups, $n = 6$). $^{\#\#}P < 0.01$, *vs* control; $^{*}P < 0.05$, *vs* CSDS. Data were analyzed by one-way analysis of variance followed by Dunnett's test. CSDS: chronic social defeat stress; FLX: fluoxetine; HE: hematoxylin-eosin; XYS: Xiaoyao San.

increasing the expression level of AdipoR1 in neurons and ultimately improve the energy metabolism balance of neurons.

4. Discussion

The adverse effects of depression on physical and mental health are not only due to changes in mental state but also due to the poor physical health that often accompanies the disorder [6]. In fact, there is strong evidence that there is a two-way link between depression and metabolic disorders [26]. Population-based cohort analyses have confirmed that type 2 diabetes and depression are indeed mutually independent risk factors [27]. Diabetic patients have an increased risk of depression, while depressed patients have an increased risk of diabetes [28]. In this study, we used CSDS, a classic modeling method, to induce depression-like behaviors in mice. The model mice exhibited reduced interaction time with CD1 mice, increased immobility time in the forced swim test, reduced latency to enter the central area in the open field test, and reduced time to enter the open arms in the elevated plus maze test. Furthermore, depression-like behavior was accompanied by an increased blood glucose level and impaired glucose tolerance during modeling. Recent studies reported that XYS is effective not only

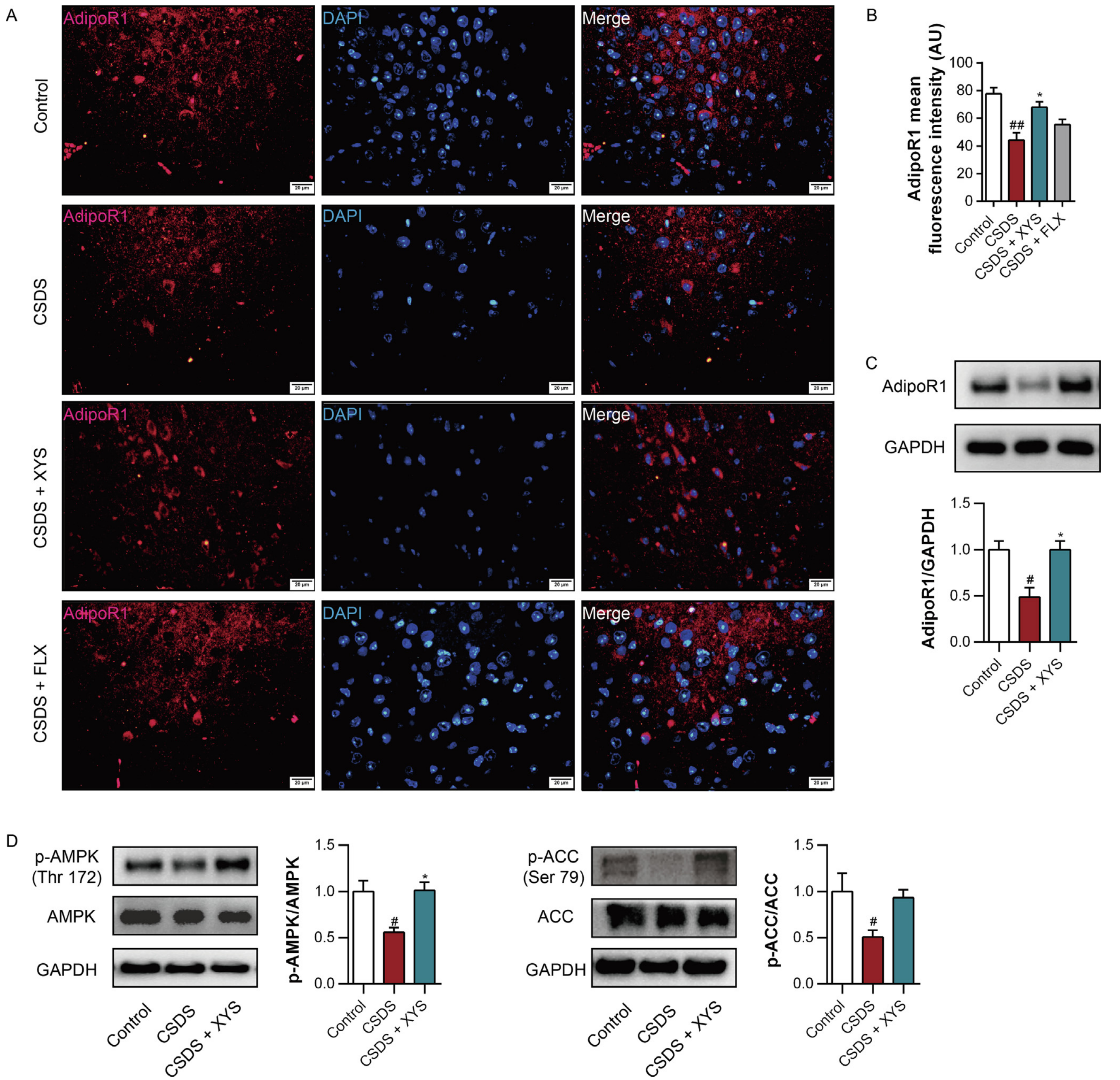

Fig. 4. Effects of XYS on the AdipoR1/AMPK/ACC pathway in the hypothalamus. (A) Representative immunofluorescence images of AdipoR1 (red) in the hypothalamic region. DAPI-stained nuclei are shown in blue. (B) Relative quantification of the intensity of AdipoR1-positive staining from immunofluorescence images in the hypothalamus region ($P = 0.0008$ within groups, $n = 4$). (C) Western blotting and semiquantitative analysis of AdipoR1 protein in the hypothalamus ($P = 0.0037$ within groups, $n = 5$). (D) Western blotting of p-AMPK, AMPK, p-ACC and ACC proteins in the hypothalamus, and semiquantitative analysis of p-AMPK/AMPK ratio ($P = 0.0060$ within groups, $n = 5$) and p-ACC/ACC ratio ($P = 0.0420$ within groups, $n = 5$). $^{\#}P < 0.05$, $^{\#\#}P < 0.01$, *vs* control; $^{*}P < 0.05$, *vs* CSDS. Data were analyzed by one-way analysis of variance followed by Dunnett's test. ACC: acetyl-coenzyme A carboxylase; AdipoR1: adiponectin receptor 1; AMPK: adenosine 5′-monophosphate-activated protein kinase; CSDS: chronic social defeat stress; DAPI: 4′,6-diamidino-2-phenylindole; FLX: fluoxetine; GAPDH: glyceraldehyde-3-phosphate dehydrogenase; p-AMPK: phospho-AMPK; XYS: Xiaoyao San.

in alleviating depression and other mood disorders but also in treating metabolic disorders such as type 2 diabetes [29,30]. In fact, we confirmed that XYS can ameliorate depression-like behavior and glucose tolerance impairment in mice. This study mainly explores the relationship between depression and metabolic disorders, including the pathogenesis of depression and the biological mechanism underlying the antidepressant effect of XYS.

Since the brain is a unique organ that uses glucose as its main source of energy, it is particularly important to ensure that enough glucose can enter the brain [31]. However, extracellular hyperglycemia is detrimental to neuronal function and can lead to chronic inflammation [32]. The hypothalamus is the main region of the central nervous system that acts as an "intersection" between depression and metabolic disorders [33]. Studies over the past 50 years have shown that hyperactivity of the HPA axis is one of the most consistent changes underlying major depression [34]. Corticotropin-releasing hormone is the basis of HPA axis activity in paraventricular nucleus neurons in the thalamus, and overactivity of corticotropin-releasing factor-expressing neurons is a prominent characteristic of depression [35]. In this study, the serum corticosterone level of mice in the CSDS group was significantly higher than that of mice in the control group, and the production of Nissl bodies in hypothalamic neurons was reduced. XYS reduced the level of corticosterone in model mice and

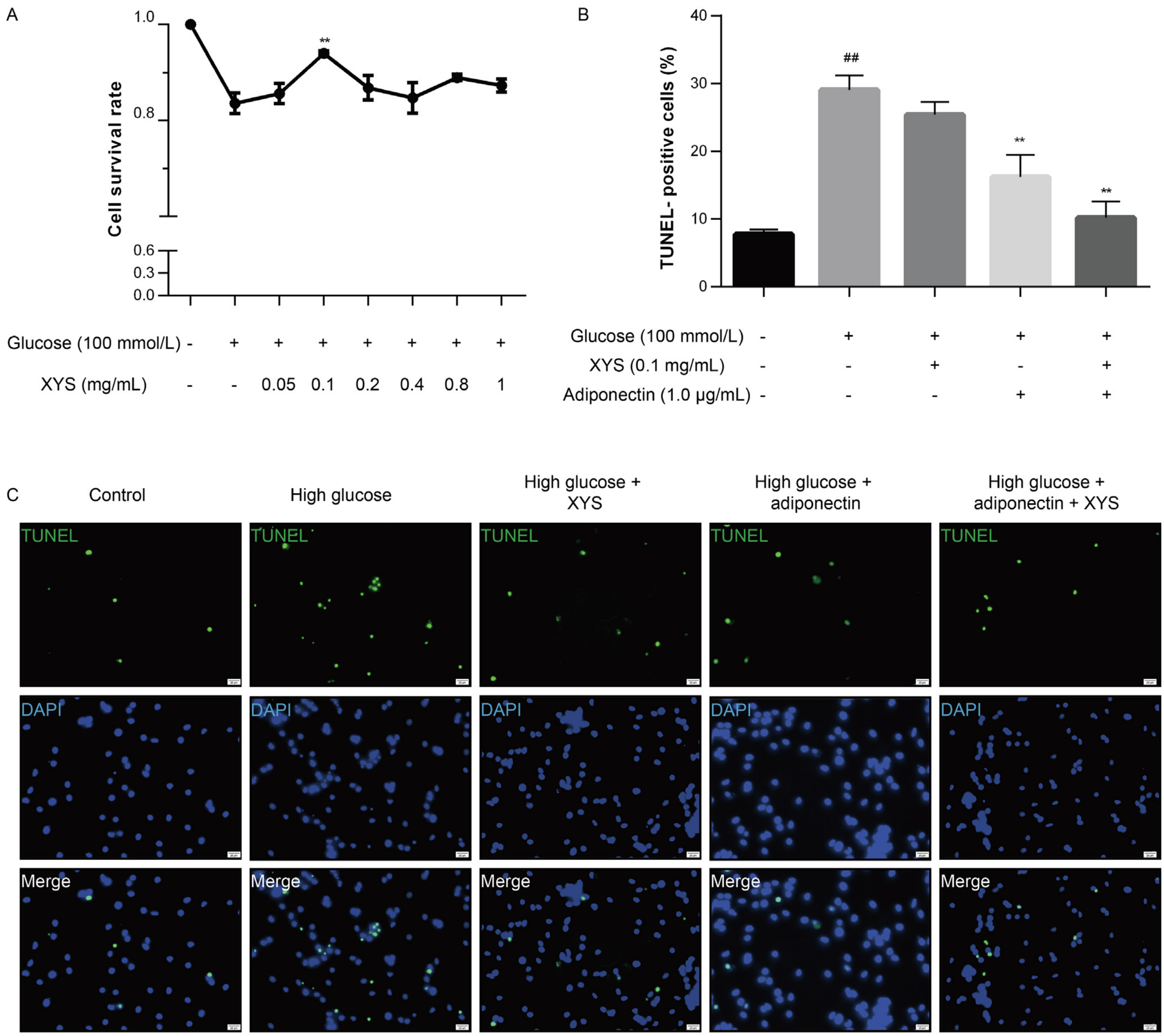

Fig. 5. XYS improved the survival rate of primary neurons and reduced the rate of apoptosis. (A) Cell survival rate after treatment with different concentrations of XYS and high glucose (100 mmol/L). (B) Percentages of apoptotic neurons evaluated by the ratio of TUNEL-positive cells ($P = 0.0001$ within groups, $n = 5$). (C) Representative photomicrographs of TUNEL staining of primary neurons. $^{\#\#}P < 0.01$, *vs* control; $^{**}P < 0.01$, *vs* high glucose. Data were analyzed by one-way analysis of variance followed by Dunnett's test. DAPI: 4′,6-diamidino-2-phenylindole; TUNEL: terminal deoxynucleotidyl transferase dUTP nick end labeling; XYS: Xiaoyao San.

increased the production of Nissl bodies in hypothalamic neurons. Regulation of HPA axis function is an important mechanism underlying the antidepressant effect of XYS.

Experimental evidence shows that adiponectin is involved in the occurrence and development of depression and acts as a "communication bridge" between depression and metabolic disorders [36,37]. Therefore, in this study, we mainly studied the effects of XYS on blood glucose levels and adiponectin levels. *In vivo*, the serum adiponectin level of mice with CSDS-induced depression-like behavior was significantly lower than that of mice in the normal control group, and XYS significantly increased the level of serum adiponectin. A study provided sufficient experimental evidence that physical exercise can regulate the secretion of adiponectin from adipocytes to regulate hippocampal neurogenesis and exert antidepressant effects. The researchers injected wild-type mice with adenovirus expressing adiponectin (Ad-Adn) or luciferase (Ad-Luc, control). Compared to Ad-Luc-treated mice, mice that received Ad-Adn injection for 2 weeks showed a significant reduction in depression-like behavior in behavioral tests [38]. In addition to exerting antidepressant effects, XYS also alleviates metabolic dysfunction, and regulates the metabolism of blood glucose and blood lipids [39]. Various active ingredients of XYS, specifically Radix Angelicae Sinensis, Radix Paeoniae Alba and Radix Glycyrrhizae, can increase serum adiponectin levels [40–42]. Thus, we speculate that the antidepressant effect of XYS may be related to an increase in serum adiponectin levels.

AdipoR1 in the hypothalamus regulates the food intake and energy balance [43]. AMPK is a recognized regulator of systemic energy balance that is activated by intracellular energy consumption caused by muscle contraction, nutrient deficiency and hypoxia and restores ATP levels by inhibiting anabolic energy consumption and promoting catabolic ATP production pathways [44,45]. As an upstream signal, adiponectin binds to AdipoR1 and stimulates the phosphorylation of AMPK to regulate cellular glucose and lipid metabolism, and energy balance [46]. In our *in vivo* experiments, we observed that the level of AdipoR1 and the phosphorylation of AMPK were increased in the hypothalamus and primary neurons in mice treated with

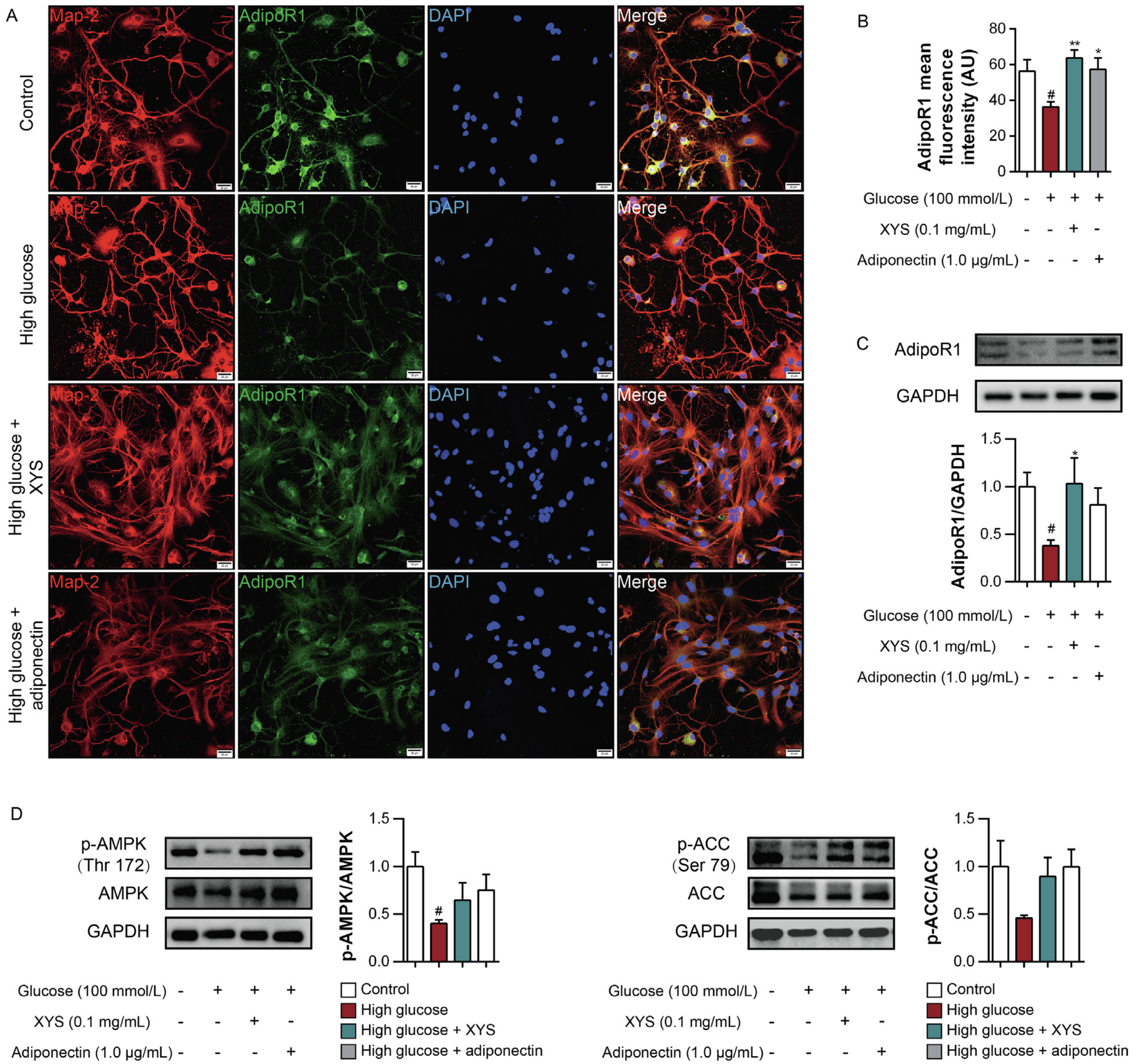

Fig. 6. Effects of XYS on the AdipoR1/AMPK/ACC pathway in primary neurons. (A) Representative immunofluorescence images of AdipoR1 (green) protein expression in primary neurons (red, Map-2 is a neuron-specific nuclear protein marker). DAPI-stained nuclei are shown in blue. (B) Relative quantification of the intensity of AdipoR1-positive staining from immunofluorescence images in primary neurons ($P = 0.0040$ within groups, $n = 12$) (twelve cells from each group were analyzed in 3 different fields of view). (C) Western blotting and semiquantitative analysis of AdipoR1 protein in primary neurons ($P = 0.1105$ within groups, $n = 3$). (D) Western blotting of p-AMPK, AMPK, p-ACC and ACC proteins in primary neurons, and semiquantitative analysis of p-AMPK/AMPK ratio ($P = 0.1003$ within groups, $n = 3$) and p-ACC/ACC ratio ($P = 0.2284$ within groups, $n = 3$). $^{\#}P < 0.05$, *vs* control; $^{*}P < 0.05$, $^{**}P < 0.01$, *vs* high glucose. Data were analyzed by one-way analysis of variance followed by Dunnett's test. ACC: acetyl-coenzyme A carboxylase; AdipoR1: adiponectin receptor 1; AMPK: adenosine 5′-monophosphate-activated protein kinase; DAPI: 4′,6-diamidino-2-phenylindole; GAPDH: glyceraldehyde-3-phosphate dehydrogenase; Map-2: microtubule-associated protein 2; p-AMPK: phospho-AMPK; XYS: Xiaoyao San.

XYS. Activation of AMPK inhibits fatty acid synthesis through phosphorylation of ACC [47]. Consistent with this hypothesis, XYS can inhibit the activity of ACC and convert ACC to p-ACC. These findings indicate that XYS effectively inhibits lipid metabolism in hypothalamic neurons and that this process mainly results from activation of AMPK and AMPK-induced phosphorylation of ACC. It is worth noting that XYS alone did not significantly alleviate the neuronal damage caused by high glucose. However, the simultaneous addition of XYS and adiponectin had a better effect than adiponectin alone. These results show that XYS strengthens the bridge between depression and metabolic disorders by increasing the expression of adiponectin receptors to increase the sensitivity of cells to adiponectin, regulating the balance of neuronal energy metabolism, and alleviating cell damage in the hypothalamus, ultimately exerting an antidepressant effect.

5. Conclusions

In general, our work provides new insights into the biological mechanism underlying the antidepressant effect of XYS. XYS can relieve HPA axis dysfunction, regulate secretion of adiponectin and metabolism, improve glucose tolerance, reduce hypothalamic cell damage, and thus alleviate depression-like behavior in mice. The antidepressant mechanism of XYS may involve regulation of the AdipoR1/AMPK/ACC signaling pathway via adiponectin. Nevertheless, how XYS participates in this pathway to ameliorate depression remains to be determined.

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Authors' contributions

J.C., X.L. and Q.M. conceived and designed the experiments; X. M. and Y.C. performed the experiments; X.Z. and D.L. analyzed the data; L.H. contributed reagents/materials/analysis tools; K.T. wrote the manuscript. All authors agree to be accountable for the content of the work.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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